

LAMONI, IA, USA

WMO: 725499

Lat: 40.631N

Lon: 93.901W

Elev: 346

StdP: 97.24

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-16.9	-23.5	0.5	-18.7	-21.3	0.6	-16.0	12.6	-2.3	11.7	-2.9	4.0	310	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	33.6	24.4	32.1	24.1	30.6	23.4	25.9	31.0	25.1	30.1	24.2	28.8	4.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	20.2	29.1	23.6	19.2	28.4	22.7	18.1	27.5	82.4	31.3	78.5	30.0	74.8	28.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.6	-23.8	36.0	3.0	1.7	-26.0	37.2	-27.7	38.1	-29.4	39.1	-31.5	40.3		
(5)			-24.2	27.4	2.9	0.6	-26.2	27.9	-27.9	28.2	-29.5	28.6	-31.6	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-4.2	-2.4	4.9	11.5	16.8	22.1	24.4	23.2	19.0	12.0	4.8	-1.7		
	DBStd	11.29	6.76	6.92	6.46	5.29	4.48	3.24	3.10	3.01	4.43	5.26	5.95	6.41		
	HDD10.0	1601	439	351	186	44	4	0	0	0	1	41	173	364		
	HDD18.3	3277	697	582	418	212	83	7	1	2	44	205	405	621		
	CDD10.0	1939	0	2	27	88	214	362	446	409	271	102	18	1		
	CDD18.3	574	0	0	1	6	36	119	188	153	64	8	0	0		
	CDH23.3	5080	0	0	7	64	272	999	1829	1295	547	67	0	0		
	CDH26.7	1768	0	0	0	10	56	319	732	467	174	10	0	0		
(14) Wind	WSAvg	4.2	4.7	4.7	4.9	5.0	4.3	3.7	3.3	3.0	3.5	4.2	4.7	4.6		
(15) Precipitation	PrecAvg	991	25	37	60	104	144	146	129	108	86	75	51	32		
	PrecMax	1438	52	101	134	206	304	339	504	242	280	158	131	98		
	PrecMin	626	6	6	3	32	48	32	50	25	14	11	1	0		
	PrecStd	218	12	22	36	39	64	69	104	65	61	40	32	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	18.5	24.2	28.0	31.1	33.5	36.6	35.9	32.8	28.0	22.4	16.1		
		MCWB	7.6	10.9	16.2	18.1	22.4	24.6	24.1	23.3	23.3	20.1	15.1	12.1		
	2%	DB	10.4	14.1	21.0	25.2	28.0	31.7	34.0	32.8	30.9	25.3	18.9	12.0		
		MCWB	6.1	9.3	13.9	16.6	20.6	23.8	24.8	23.9	22.9	18.3	13.7	8.9		
	5%	DB	7.5	10.6	17.9	22.7	26.4	30.0	32.4	31.2	28.7	22.8	16.3	9.2		
		MCWB	4.2	6.9	12.6	15.4	19.5	22.9	24.7	23.7	21.5	16.6	11.6	6.4		
	10%	DB	4.7	7.3	15.1	20.4	24.5	28.4	30.9	29.1	26.5	20.7	13.7	7.1		
		MCWB	2.3	4.2	10.7	14.3	18.5	22.1	24.0	22.9	20.2	15.8	9.8	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	13.3	17.5	19.7	23.8	26.2	27.4	26.9	25.0	22.1	16.5	13.4		
		MCDB	11.8	15.7	22.3	25.9	29.6	31.6	32.3	31.4	30.6	26.2	19.9	15.3		
	2%	WB	6.4	9.7	15.6	18.0	22.0	24.8	26.2	25.6	23.8	20.1	14.5	9.8		
		MCDB	9.5	14.2	18.9	23.3	26.8	29.9	31.4	30.3	28.9	23.0	17.8	10.9		
	5%	WB	4.5	7.0	13.7	16.5	20.6	23.8	25.5	24.7	22.7	18.0	12.5	6.8		
		MCDB	7.7	10.2	17.5	21.5	25.2	28.6	30.6	29.2	27.3	21.0	15.6	9.0		
	10%	WB	2.4	4.1	10.7	15.0	19.4	22.9	24.6	23.7	21.6	16.1	10.3	4.5		
		MCDB	4.6	7.1	14.9	19.4	23.5	27.5	29.2	28.0	25.3	19.9	13.4	6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.9	10.6	11.2	10.4	10.2	10.3	10.3	11.3	10.8	10.0	8.9		
		MCDBR	12.9	14.1	14.0	14.4	11.9	11.4	11.8	12.3	12.4	13.1	13.2	12.0		
	5% WB	MCWBR	9.3	9.8	8.2	8.4	6.5	5.6	4.9	5.4	5.6	7.3	8.5	9.1		
		MCWBR	12.2	13.1	12.1	12.7	10.6	10.5	10.4	10.6	10.8	10.6	11.8	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.316	0.331	0.386	0.411	0.419	0.429	0.423	0.379	0.335	0.296	0.284		
	taud		2.438	2.361	2.402	2.264	2.260	2.279	2.247	2.285	2.357	2.475	2.522	2.471		
	Ebn at Noon		878	896	922	886	868	859	845	840	858	865	865	854		
	Edh at Noon		78	99	106	131	134	132	135	127	110	88	72	70		
(44) All-Sky Solar Radiation	RadAvg		2.00	2.92	3.79	4.82	5.69	6.31	6.34	5.53	4.71	3.20	2.20	1.71		
	RadStd		0.19	0.21	0.32	0.32	0.47	0.57	0.47	0.32	0.31	0.36	0.19	0.18		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page